Identification of "Viral Travel" Destinations and Social Media Engagement Factors Relevant to Lufthansa Flight Booking Surge Prediction

Sponsor: Lufthansa Group

Team 4

Country names have been redacted at the sponsor's request.

I. Introduction

The Lufthansa Group is an aviation group with operations worldwide and is particularly dominant in the European market. With the recent rise of "viral travel," a phenomenon in which travel-related social media content influences how and where people travel, the Lufthansa Group has an opportunity to enhance their strategic planning by identifying the next viral travel destinations and predicting future flight booking surges.

This project aims to transform social media data into a relevance score to predict the "virality" of European travel destinations. Lufthansa Group flight booking data from 2024 and 2025 is cross-referenced with scraped YouTube and TikTok travel video data to identify emerging viral travel destinations and the social media engagement factors relevant to predicting flight booking surges.

The goal is to aid the Lufthansa Group in making informed business decisions related to flight route planning, resource allocation, and ticket pricing based on information extracted from travel-related social media posts.

II. Data Collection and Preprocessing

A. Lufthansa Group Flight Booking Data

The Lufthansa Group provided flight booking transaction data for the period from 12/9/2024 to 12/9/2025 for 222 European airports. The data contains information including booking date, travel date, departure airport, arrival airport, and number of passengers on the booking.

Minimal data preprocessing was required, including rounding up decimal passenger values, merging on the airport city and country, and limiting to bookings with passenger values greater than zero.

Preliminary exploratory data analysis revealed distinct, high-traffic corridors with routes departing from [COUNTRY], [COUNTRY], and [COUNTRY] and arriving in destinations such as [COUNTRY], [COUNTRY], and [COUNTRY]. While flight bookings generally have a seasonal component, the data provided contains only one period (one year). Thus, models such as SARIMA and Holt-Winters are not suitable given the data limitations.

B. Social Media Data

Social media data was gathered via official APIs and data scraping techniques. Applications for the Meta API (Instagram, Facebook) and Reddit API were submitted; the Meta application did not move past the "review" stage and the Reddit application was rejected. Efforts were therefore focused on YouTube data via Google Cloud API and TikTok data via Selenium.

YouTube Data. YouTube video data was gathered in three steps. First, the first 50 pages of YouTube search results were obtained for each country for the period from 6/1/2024 to 12/9/2025, using the country name with a "travel" addon as the query (e.g., "Travel Germany"). Due to time and API token limitations, data were collected at the country level rather than city level. Overall, 27,495 unique video IDs were gathered for 42 countries.

Second, the video IDs were used to gather metadata including title, channel ID, channel name, publish date, description, keywords, total views, total likes, total comments, and total favorites.

Third, the top 20 comments per video were gathered, yielding 144,432 total comments (≈3,900 per country on average). The top comments serve as a proxy for overall viewer sentiment.

TikTok Data. TikTok video data was gathered via Selenium. We scraped individual hashtags on TikTok, gathering similar data about viewer metrics, descriptions, and comments. Due to the small number of videos per hashtag, we scraped multiple hashtags per country, such as "#travelspain" and "#discoverspain." Due to factors observed during the scraping process (namely that hashtags related to the UK provided less data than those relating to its constituent countries, and that hashtags related to Georgia almost universally pertained to the American state rather than the European country), we scraped hashtags pertaining to Wales, Scotland and England rather than those pertaining to the UK, and we omitted hashtags pertaining to Georgia. In total, 14,090 TikTtok videos were gathered.

III. Analytical Approach

The analytical approach consists of three main parts:

1. A multi-step approach on flight booking data to identify booking surges (smoothing, normalization, clustering, surge detection)
2. Sentiment analysis on social media comments and summarization of engagement factors

3. Merging booking surges with social media data and training an XGBoost model to predict viral travel destinations and identify relevant features

Two XGBoost models were trained and tested. Model 1 correlates each country's total weekly flight booking passengers with the calculated Relevance Score to identify predictive countries, where social media data leads flight booking trends, and reactive countries, where social media data follows flight booking trends. In addition to identifying countries as predictive or reactive, Model 1 also provides insight on the number of weeks of the lead or lag for each country. Model 2 estimates the probability that a country will see a spike in flight bookings within a specified time horizon from its social media data spikes. Model 2 also provides insight into the importance of the features included in the XGBoost model.

A. Flight Booking Data Surge Detection

1. Convert flight bookings to z-scores. The flight booking data was converted to z-scores to standardize across all airports, allowing airports with vastly different booking volumes to be compared.

2. Smooth using exponential weighting. Each airport's z-scores were smoothed using Exponential Weighted Moving Average (EWM) to reduce noise. An alpha value of 0.25 was used to balance short-term responsiveness and long-term stability. Additional alpha values of 0.01, 0.05, and 1 were also tested.

3. K-means clustering. Using the smoothed z-scores, k-means clustering was performed to segment like-airports into groups, enabling individual airports to be compared to their cluster. Euclidean distance was used and 4 clusters were identified. Additional runs with 1, 10, and 20 clusters were tested.

4. Calculate residuals. Within each cluster, flight booking residuals were calculated by comparing each airport's smoothed z-score to the cluster's trimmed mean to understand each airport's over- or under-performance relative to its cluster.

5. CUSUM on residuals. A Cumulative Sum (CUSUM) control chart was run on the residuals to detect spikes in the booking data.

B. XGBoost Flight Booking Surge Prediction: Model 1

The booking surge signals and social media engagement features were merged and an XGBoost model was trained and tested to calculate the social media trending score and relevance score, which then used to identify if social media surge leads the booking surge.

The process is outlined below:

1. Social Media Comment Sentiment Analysis. Sentiment analysis was performed on social media comment data from both TikTok and YouTube to quantify audience reactions to travel content. Comment-level sentiment scores were aggregated to the post/video level and then to a weekly country-level signal used as a feature in the trending score models.

a. TikTok Sentiment Analysis. TikTok comment sentiment was computed using the **cardiffnlp/twitter-roberta-base-sentiment-latest** transformer model — a RoBERTa-based model fine-tuned on Twitter data and well-suited to the short, informal language typical of TikTok comments. For each post, the comments were passed through the model in batches. The model returns three class probabilities (positive, neutral, negative); a continuous compound score is derived as:

$$\text{score} = P(\text{positive}) - P(\text{negative}) \in [-1, +1]$$

The **median** compound score across all comments for a post is stored as `comment_median_sentiment_score`. Posts with no comments receive a score of 0. The pipeline processed **19,941 posts** across 43 countries.

Example. A post about Copenhagen with comments such as *"This city is absolutely stunning!"* yields a score near $+1$. A post with mixed or neutral comments scores near 0.

b. YouTube Sentiment Analysis. YouTube comment sentiment followed the same transformer-based approach. For each video, the top 20 comments were retrieved via the Google Cloud API and passed through the same model in batches of 10. The median compound score per video was stored as `comment_median_sentiment_score`.

The pipeline processed **27,495 videos** across 42 countries. After joining video metadata with comment scores and removing videos with no comments, **21,460 videos** with valid sentiment scores were retained.

2. Aggregation to Weekly Country-Level Signal. Both TikTok and YouTube sentiment scores were aggregated from the post/video level to a weekly country-level signal by computing the **median sentiment score** across all posts or videos for a given country and week. This weekly sentiment signal was used as a feature in the XGBoost trending score models and as a component of the Relevance Score formula.

3. Build Weekly Social Trending Scores.

a. TikTok Trending Score. An XGBoost regression model is trained on post-level features aggregated to weekly country-level observations. The target is total weekly engagement (likes + comments + bookmarks). To prevent target leakage, these three engagement components are *removed* from the feature set; only `post_count` and `sentiment` are used as current-week inputs.

The model output is min-max normalized per country to a 0–100 scale. A CUSUM signal is then computed on this score to detect sustained upward momentum:

$$s_{\text{hi}}[i] = \max\big(0,\ s_{\text{hi}}[i-1] + z_i - k\big), \quad z_i = \frac{x_i - \mu}{\sigma}$$

where $k = 0.75$ is the slack parameter and the alarm fires when $s_{hi} > h = 5.0$. The TikTok dataset covers 2,925 country-week observations (2024-06-17 to 2025-12-29) with a signal rate of 5.4%.

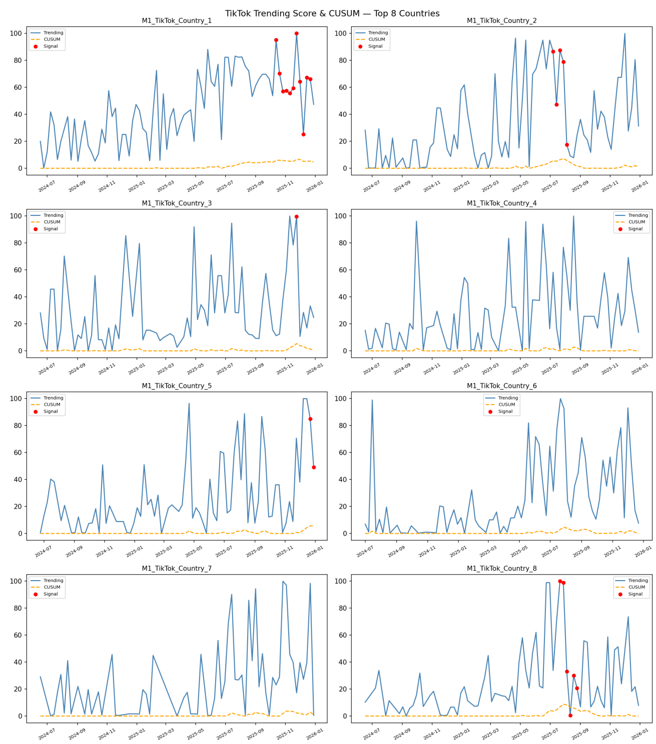

Figure 1: Weekly TikTok CUSUM for the top 8 countries. Orange dashed line = CUSUM score; red dots = alarm weeks where the score exceeds the threshold $h = 5.0$.

b. YouTube Trending Score. An identical pipeline is applied to YouTube data (3,252 observations, signal rate 5.3%). The same CUSUM parameters ($k = 0.75$, $h = 5.0$) are used. YouTube's top feature is `video_count` (20.2% importance), lower than TikTok's `post_count` (32.5%), reflecting the more complex engagement dynamics of longer-form video content.

4. Compute the Relevance Score. The two social signals and the booking CUSUM are fused into a single weekly **Relevance Score** (0–100) per country using weights derived from XGBoost feature importance:

$$\mathrm{RS} = 0.166 \cdot \mathrm{BK}_{\mathrm{norm}} + 0.056 \cdot \overline{\mathrm{TT}} + 0.436 \cdot \overline{\mathrm{YT}}$$

where $\overline{\mathrm{TT}} = 0.5 \cdot \text{TikTok\_Score} + 0.5 \cdot \text{TikTok\_CUSUM}$ and similarly for YouTube. The score is then re-normalised 0–100 per country to remove country-size effects. The fusion model achieves Test $R^2 = 0.884$.

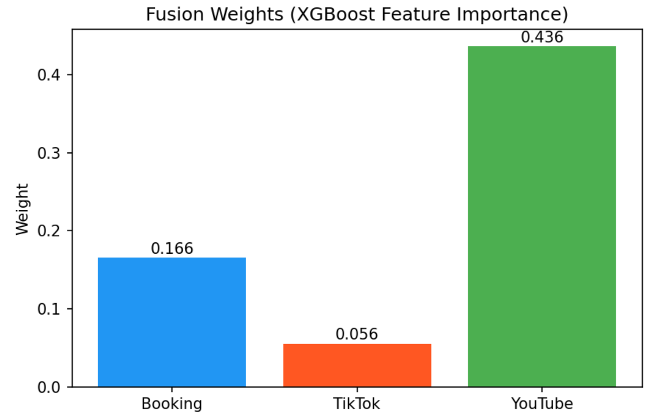

Figure 2: Fusion weights derived from XGBoost feature importance. YouTube dominates (0.436) because its longer content lifecycle produces more sustained, predictive signals than TikTok (0.056). Booking CUSUM contributes 0.166 as a ground-truth demand anchor.

How to read the Relevance Score. Figure 3 shows Model 1 Sample Country 1 (M1SC1)'s weekly signals. When TikTok, YouTube, and Booking CUSUM all rise simultaneously, the purple Relevance Score peaks. When only one platform spikes, the score stays moderate — a single-platform signal is noisy and unreliable.

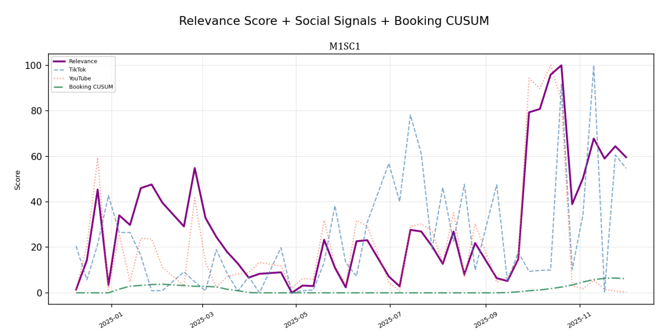

Figure 3: [M1SC1] Relevance Score (purple) overlaid with TikTok (blue dashed), YouTube (red dotted), and Booking CUSUM (green dash-dot). The October 2025 peak (score = 100) is driven by simultaneous TikTok and YouTube spikes. The February 2025 spike is YouTube-only, producing a moderate but not maximum score.

5. Lead-Lag Cross-Correlation. For each country, the Pearson correlation between the weekly Relevance Score and weekly booking passengers is computed at lags $\ell \in [-8, +8]$ weeks. The lag that maximises correlation defines the country's relationship:

- $\ell < 0$: social media *leads* bookings — **predictive**
- $\ell > 0$: bookings lead social — **reactive**

Example. For [M1SC1] ($\ell = -6$, $r = 0.628$), the correlation peaks when the social signal is shifted 6 weeks *earlier* than the booking signal, meaning social content about [M1SC1] consistently appears 6 weeks before passengers book flights there.

Across 41 countries, **12 are classified as predictive** (social leads bookings) and 29 as reactive.

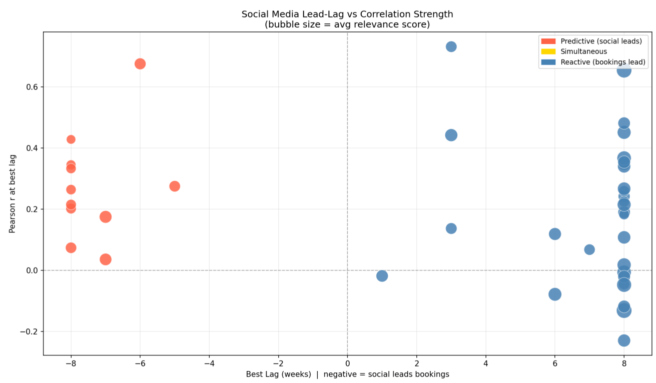

Figure 4: Pearson r vs. optimal lag for all 41 countries. Red points = predictive (social leads bookings); blue = reactive. [M1SC1] (lag $= -6$, $r = 0.628$) is the top predictive country. [M1SC2] (lag $= +3$, $r = 0.459$) is reactive despite high correlation.

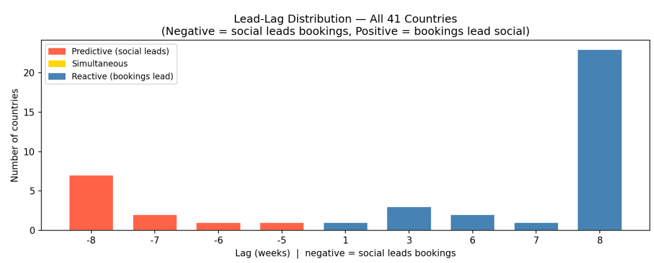

Figure 5: Distribution of optimal lag values. The majority of countries cluster at $\ell = -8$ weeks (social leads by 8 weeks) or $\ell = +8$ weeks (bookings lead by 8 weeks).

Important caveat. A high Pearson r at a negative lag is a *hypothesis*, not a confirmation. Validation against discrete spike-to-surge events (Step 4) is required before any country is designated as a genuine leading indicator.

6. Spike-to-Surge Validation.

a. Defining spikes and surges.

- **Social spike:** a week where the combined social signal $(\text{TikTok} + \text{YouTube})/2$ exceeds the country's 75th percentile threshold.
- **Booking surge:** a week where the booking CUSUM signal fires (i.e., $s_{\text{hi}} > 5.0$ on normalized weekly passengers).

b. Matching logic. For each social spike week W in a predictive country with lag ℓ, the model predicts a booking surge in the window $[W + |\ell| - 2,\ W + |\ell| + 2]$ (± 2 week tolerance). A spike is *confirmed* if a booking surge actually occurs in that window.

c. Metrics.

$$\text{Precision} = \frac{\text{confirmed spikes}}{\text{total spikes}}$$

$$\text{Recall} = \frac{\text{confirmed spikes}}{\text{total surges}}$$

$$F_1 = \frac{2 \cdot P \cdot R}{P + R}$$

Table 1: Predictive vs. Reactive Country Validation

Group	Precision	Recall	F1
Predictive (12 countries)	0.100	0.348	0.153
Reactive (29 countries)	0.171	0.320	0.213
Lift (pred / react)	**0.59**$\times$	**1.09**$\times$	—

d. Aggregate results. Reactive countries show higher average Precision and F1 due to strong seasonal countries ([M1SC2], [COUNTRY], [COUNTRY]). However, reactive countries *cannot* be used for proactive marketing — their social signals follow bookings rather than precede them.

Table 2: Top 5 Predictive Countries by F1 Score

Country	Lag	r	P	R	F_1	Conf.
1. [M1SC1]	-6w	0.628	0.333	1.000	0.500	4/12
2. [COUNTRY]	-7w	0.039	0.273	0.643	0.383	3/11
3. [COUNTRY]	-8w	0.091	0.250	0.700	0.368	3/12
4. [COUNTRY]	-8w	0.328	0.167	0.750	0.273	2/12
5. [COUNTRY]	-8w	0.349	0.091	0.625	0.159	1/11

e. Top validated predictive countries. [M1SC1] achieves **perfect Recall (1.000)**, meaning every booking surge was preceded by a social spike. The Precision of 0.333 means one-third of social spikes led to confirmed surges.

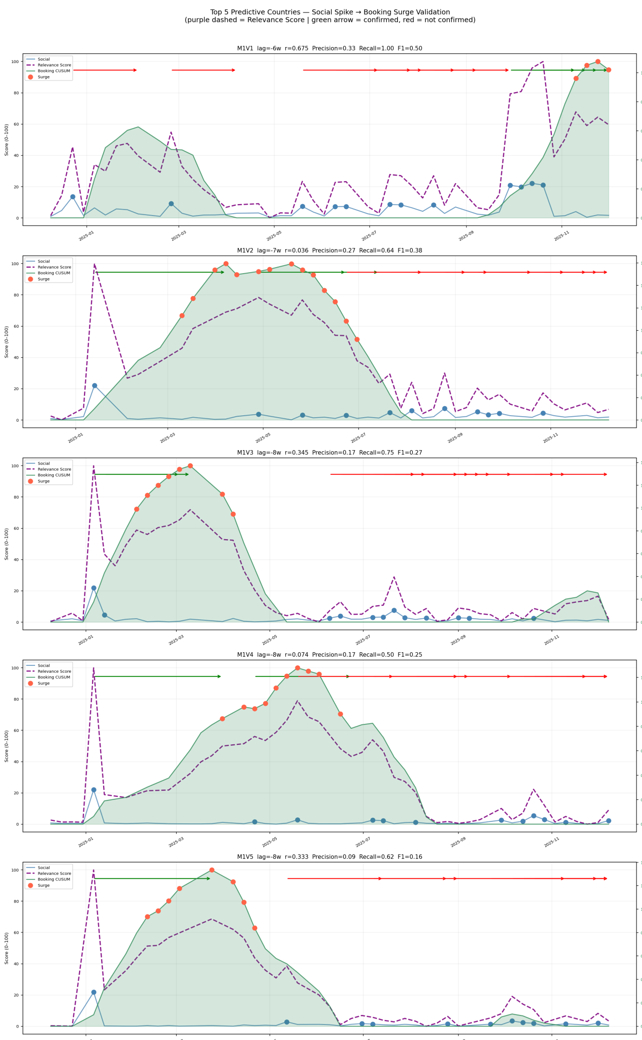

Figure 6: Spike-to-surge validation for the top 5 predictive countries. Each panel overlays the social signal (blue), Relevance Score (purple dashed), and actual passengers (green). Green arrows = confirmed predictions; red arrows = false alarms.

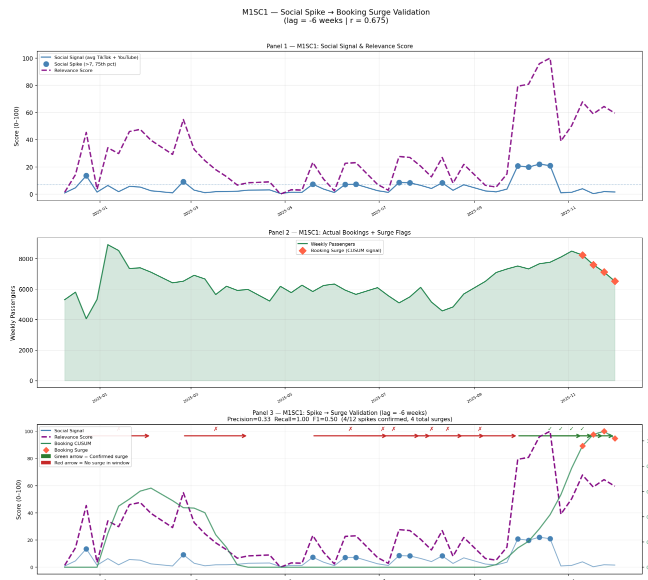

Figure 7: [M1SC1] validation detail. **Panel 1:** social signal (blue) and Relevance Score (purple dashed) with spike markers. **Panel 2:** actual weekly passengers with surge markers (red diamonds). **Panel 3:** prediction test — each arrow connects a social spike to its expected surge 6 weeks later. Green = confirmed; red = false alarm. 4 of 12 spikes confirmed (P=0.333, R=1.000, F1=0.500).

7. Detailed Country Walkthrough.

a. [M1SC1] (F1 = 0.500, lag = -6 weeks). The Model 1 Sample Country 1 (M1SC1) is the best validated predictive country. Of 12 social spikes, 4 were confirmed by booking surges within the 6$\pm$2 week window.

Concrete example. On 29 September 2025, [M1SC1]'s social score spiked above the 75th percentile threshold. The model predicted a booking surge around 10 November 2025 (6 weeks later). A booking surge was confirmed in that window — Lufthansa had a 6-week window to activate a targeted [M1SC1] campaign.

b. [M1SC2]: High F1 Score but Reactive (lag = $+3$ weeks). Model 1 Sample Country 2 (M1SC2) achieves the highest F1 Score of any country (0.909, 10 of 12 spikes confirmed) but is classified as **reactive** — bookings lead social media by 3 weeks. [M1SC2]'s social media activity responds to booking demand rather than predicting it, making it unsuitable for proactive campaign activation.

8. Key Takeaway: Directionality Over F1 Score. The most important lesson from this analysis is that **F1 Score alone does not determine actionability**. [M1SC2]'s F1=0.909 is the highest in the dataset, yet it is commercially irrelevant for advance campaign planning because its lag is positive.

The decision rule for Lufthansa is:

1. **Negative lag required:** social media must precede bookings
2. **F1 $\geq$ 0.20:** sufficient validation evidence
3. **Act on every spike:** for countries meeting both criteria, treat every social spike as a campaign trigger

Under this rule, **[M1SC1]** (lag $= -6$w, F1 $= 0.500$) is the primary actionable market, followed by [COUNTRY] (lag $= -7$w, F1 $= 0.383$) and [COUNTRY] (lag $= -8$w, F1 $= 0.368$).

C. XGBoost Flight Booking Surge Prediction: Model 2

The booking surge signals and social media engagement features were merged and an XGBoost model was trained and tested to predict the probability that a country will see a spike in flight bookings within a specified time horizon.

The process is outlined below:

1. Build Social Media Dataframes. A comprehensive social media data frame was created including the following elements:

1. **YouTube Engagement Factors:** views, likes, comments
2. **TikTok Engagement Factors:** likes, comments, bookmarks
3. **Named Entity Recognition:** relevant terms from NER
4. **NER metrics**: total term count, number of nonzero entities, highest number of counts per any one term, and term concentration (ratio of highest term to total term count)
5. **Hashtags:** hashtags from video descriptions
6. **Temporal features:** month, week of year, and quarter

The engagement factor data was available in the scraped YouTube and TikTok data. From the video descriptions, NER analysis was performed to identify terms and hashtags present in video descriptions; Claude was subsequently used to narrow down these terms and hashtags to ones that would be relevant to European travel media. Counts were created for the relevant terms and binary variables were created for hashtags, in order to count the number of times terms appeared and the number of videos that used specific hashtags.

For each airport and each day, the previous 90 days of social media data was consolidated for the country the airport is located in. This daily, airport-level data was then merged to the flight booking CUSUM data.

It should be noted that during this process, TikTok videos pertaining to Scotland, England, and Wales were relabeled as pertaining to the United Kingdom, and videos pertaining to Georgia were dropped.

2. Consolidate into Weekly, Country-Level Data. The airport-level data is then consolidated into weekly, country-level data to align with the scraped social media data which was obtained using country-specific searches. Spikes in the flight booking data detected for any airport within a country would be considered a spike for the country. After aggregation, the number of airports in the country was added to the dataframe.

3. CUSUM Time Horizon Analysis. A time horizon analysis was conducted to identify CUSUM flight booking spikes within a specified period of time. Time horizons of 1 week, 1 month, and 3 months were tested, and the 3 month time horizon yielded the best result. Thus, a 3-month time horizon identifier was created by assigning a value of 1 to all weeks within 3 months of an observed CUSUM booking spike.

4. Train and Test XGBoost Model. An XGBoost model was trained and tested on the weekly, country-level data using leave one out cross validation where an entire country was left out for each run. The XGBoost model output is the predicted probability that a country's flight booking data will see a spike.

5. Calculate Relevance Score. The Relevance Score was calculated using the probability of a flight booking spike within the specified time horizon. A threshold of 0.5 was used to indicate instances where a flight booking spike was predicted to occur.

6. Overall Feature Importance. In addition to the predicted spikes, Model 2 also identifies the feature importance for each country and for the model overall. Figure 8 below shows the top 20 features with the highest importance (left) and the distribution of feature importance (right) for the overall model. The top 20 features include the number of airports in the country and many of the relevant terms from the NER analysis such as "Reykjavik," "Oslo," "Cyprus," "Belgium," and "Austria." It also includes three hashtags including "#traveleurope," "#icelandtravel," and "#walkingtour." Notably, none of the social media engagement factors are in the top 20.

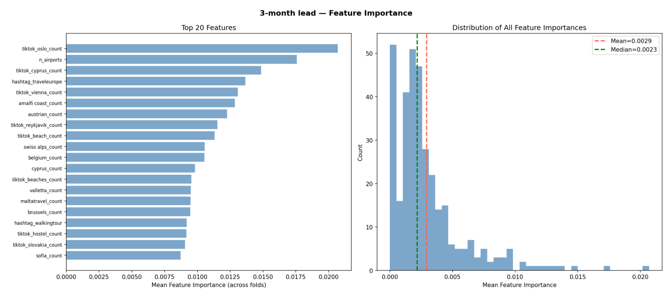

Figure 8: Model 2 overall feature importance.

7. Sample country-level statistics.

a. [M2SC] Overview. The sample analysis for Model 2 Sample Country (M2SC) shows an example of a true positive and a false positive. In Figure 9 below, the blue line represents the predicted probability of a flight booking spike occurring where values above 0.5 are instances where a spike is predicted. The blue shaded regions represent the 3-month horizon from each predicted spike. The orange line represents an occurrence of a detected CUSUM spike based on the flight booking data.

Figure 9 shows two distinct regions with predicted flight booking spike horizons. There was a CUSUM booking spike within the first shaded region, depicted by the orange vertical line, which is a true positive prediction. The second shaded region did not have a CUSUM booking spike which is a false positive prediction.

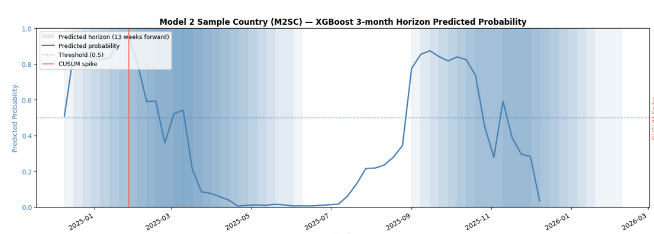

Figure 9: Model 2 analysis for [M2SC].

b. [M2SC] Feature Importance. Figure 10 below shows the feature importance for [M2SC]. Overall, the top 20 features included the number of airports in the country and many relevant terms from the NER analysis such as “Slovakia,” “Vilnius,” and “Beach.” Notably, the most important features were not directly related to [M2SC]; many of them instead pertained to Eastern European or Mediterranean-adjacent places (such as “Riga” and “Drubovnik”), or features present in the country, such as the Alps. This may suggest that the spike in bookings to [M2SC] were not driven by interest in the country itself, but instead by an interest in similar nations or natural features present in nearby countries. Social media engagement factors such as video views, comments, likes, and bookmarks did not make the top 20.

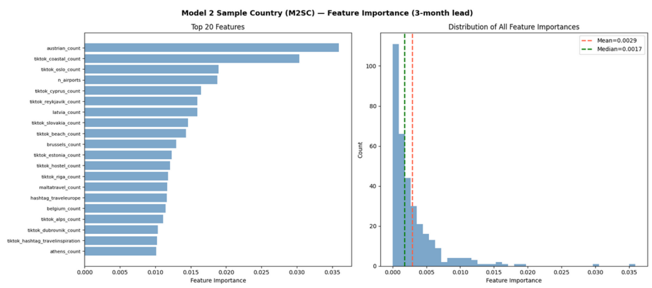

Figure 10: Model 2 feature importance for [M2SC].

IV. Evaluation:

A. Model 1

Model 1 performance was evaluated using standard regression and classification metrics including R^2, RMSE, Precision, Recall, and F1 Score. The lead-lag relationship between social media signals and booking surges was validated by checking whether social media spikes preceded booking surges within the predicted lag window (± 2 week tolerance).

Key findings:

- **[M1SC1]** (lag $= -6$ weeks, F1 $= 0.500$) is the best validated predictive country — social media activity precedes bookings by 6 weeks with perfect Recall.
- **[COUNTRY]** (lag $= -7$ weeks, F1 $= 0.383$) and **[COUNTRY]** (lag $= -8$ weeks, F1 $= 0.368$) are the next best validated predictive countries.
- **[M1SC2]** (lag $= +3$ weeks, F1 $= 0.909$) achieves the highest F1 overall but is *reactive* — bookings lead social media, not the other way around.
- The Relevance Score fusion model achieved **Test** $R^2 = 0.884$, confirming that combining TikTok, YouTube, and booking signals substantially improves predictive power.

B. Model 2

Model 2 performance was evaluated using Precision, Recall, and F1 Score for the predicted flight booking spikes using a 3-month time horizon. The confusion matrix for Model 2 is below in Table 3 and the evaluation metrics are in Table 4.

Table 3: Confusion Matrix

	Predicted Positive	Predicted Negative
Actual Positive	308	353
Actual Negative	157	1,355

Table 4: Model Performance Metrics

Metric	Value
Precision	0.662
Recall	0.466
F1 Score	0.547

Overall, Model 2 achieved a Precision of 0.662, indicating that the predicted flight booking spikes were generally reliable with a low rate of false positives. The Recall was 0.466 which indicates that the model is identifying less than half of the true booking surges and is missing a large portion of the booking surges. Finally, the F1 Score was 0.547 which indicates that the model tends towards prediction accuracy rather than prediction of all booking surges.

While the Recall and F1 Score were lower, a higher Precision is a positive takeaway for the Lufthansa Group marketing efforts. The model’s propensity to predict booking surges conservatively allows the marketing efforts to be more tailored to instances where a booking surge is likely to occur.

V. Insights and Recommendations to The Lufthansa Group

A. Model 1

Based on the analysis from Model 1, the following actions are recommended:

Table 5: Prioritized Marketing Activation Recommendations

Priority	Country	Action	Lead
1	[M1SC1]	Activate targeted digital campaigns on every social spike	6 wk
2	[COUNTRY]	Monitor social spikes; prepare campaign assets	7 wk
3	[COUNTRY]	Monitor — moderate signal	8 wk
4	[COUNTRY]	Monitor — longer lead time	8 wk
5	[M1SC2]	Seasonal monitoring only (reactive country)	Seasonal

The Relevance Score (0–100) provides a weekly, per-country composite signal that Lufthansa can use to prioritize marketing spend. Countries with rising Relevance Scores and negative lead-lags represent the highest-confidence opportunities for proactive campaign activation.

B. Model 2

Based on the analysis from Model 2, we noted that the terms and hashtags used in the video descriptions of videos pertaining to a country matter more than the specific metrics (such as likes and views) of said videos. Additionally, we found that terms relating to natural features shared by multiple countries, as well as those pertaining to similar countries, could help drive interest in less-popular tourist destinations such as [M2SC]. Given these findings, we recommend that the Lufthansa Group focuses future social media data analytics efforts on analyzing a vast array of content rather than focusing on content with certain levels of engagement.

VI. Future Analysis

A. Analysis of Additional Time Periods

Future analysis with additional data would be beneficial to more accurately identify long-term trends and seasonality. By analyzing more flight data (i.e., a longer time period), the analytical methods would have more information to train on and would likely result in more accurate and informative predictions.

Additional data with at least 2–3 periods (years) would allow for a seasonal component to be incorporated into the models. A longer time horizon would also provide more benchmarks of historical booking data and indicate how trends have changed over time.

Similarly, a longer period of social media data would show historical social media engagements for each country. Given the resources required to book a vacation, there is likely a longer lag between the rise of a new viral travel destination and flight booking surges for common travelers. A longer time horizon would allow for flight booking surge analysis for various traveler segments: early adopters, early majority, late majority, and laggards.

B. City-Specific Analysis

The present analysis leverages country-specific social media data for efficiency. Future analysis could include gathering social media data using city-specific searches. In doing so, city-level social media data could be mapped to flight booking data based on airport proximity, resulting in more accurate identification of specific airports expected to see a booking surge.

This analysis would require additional time and data scraping resources to obtain social media search results for hundreds of European cities as opposed to the 41 European countries searched in the present analysis.

C. Large-Scale Data Scraping for Businesses

The main bottleneck in this project was the time and resources required for social media data scraping. This included applying for API access as university students and utilizing Google Cloud API and Selenium when official API access was not approved. Our YouTube data scrape was limited to Google Cloud's daily token allocation, which allowed us to scrape only a few countries per day.

However, the Lufthansa Group has alternative options as a large corporation. Although it would require an investment, the Lufthansa Group could explore business-tier APIs to gather large quantities of data very quickly, significantly accelerating the data collection pipeline.

VII. Conclusion

Overall, this project successfully leveraged scraped social media travel post data to inform patterns identified in the Lufthansa flight booking data. The pipeline transforms raw TikTok and YouTube engagement data into a weekly Relevance Score (0–100) per country, identifies countries where social media activity leads booking demand by 5–8 weeks, and provides a validated early-warning system for proactive marketing activation.

The Model 1 headline finding is that **[M1SC1]** is the most consistently validated predictive country across both pipeline configurations, with a 6-week advance warning window and perfect Recall — every booking surge was preceded by a social media spike. This gives Lufthansa a reliable, actionable signal with sufficient lead time for campaign planning, creative production, and media buying.

In the meantime, Model 2 provides both a template model for predicting spikes in viral tourism using features taken from social media video descriptions and an insight into how social media interest in certain tourist destinations may drive tourism towards more off-the-beaten-path destinations.

VIII. Statement of AI Usage

The author acknowledges the use of AI tools, specifically Gemini and Claude Code, to assist in the technical development and documentation of this practicum project. The AI was utilized as a collaborative assistant in the following capacities:

- **Code Development and Debugging:** Assisted in identifying logic errors, resolving runtime exceptions, and optimizing Python scripts for data analysis and improving efficiency.
- **Methodological Support:** Provided suggestions for analytical frameworks and statistical methods appropriate for the project requirements.
- **Data Visualization:** Helped in generating Matplotlib code structures for creating figures and ensuring graphical consistency.
- **Technical Documentation:** Facilitated the resolution of LaTeX compilation errors and assisted in formatting complex document structures.

All AI-generated suggestions were manually reviewed, verified for technical accuracy, and refined by the author to ensure they met the project's academic and professional standards.

IX. Appendix: Expanded Figures

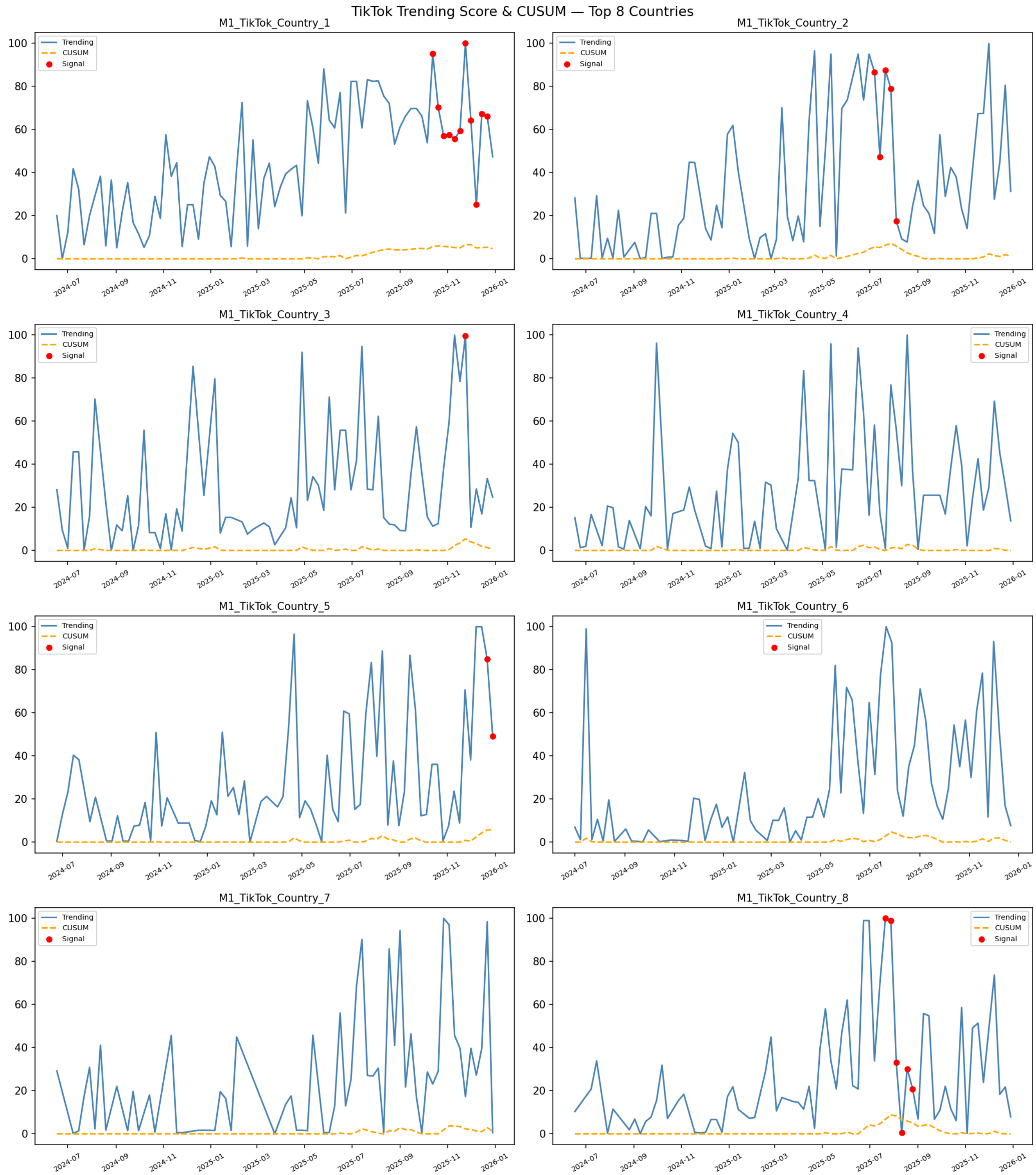

Figure 1: Weekly TikTok CUSUM for the top 8 countries. Orange dashed line = CUSUM score; red dots = alarm weeks where the score exceeds the threshold $h = 5.0$.

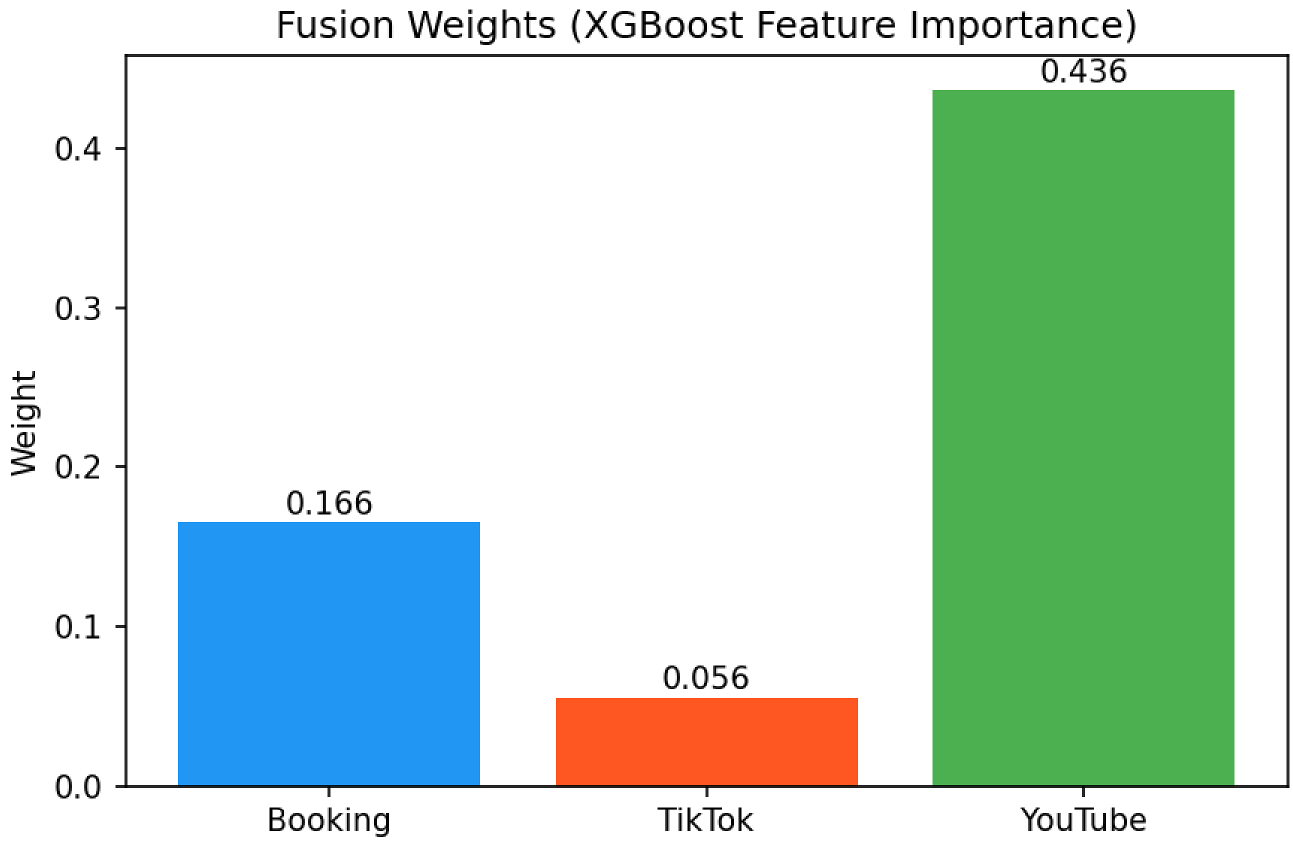

Figure 2: Fusion weights derived from XGBoost feature importance. YouTube dominates (0.436) because its longer content lifecycle produces more sustained, predictive signals than TikTok (0.056). Booking CUSUM contributes 0.166 as a ground-truth demand anchor.

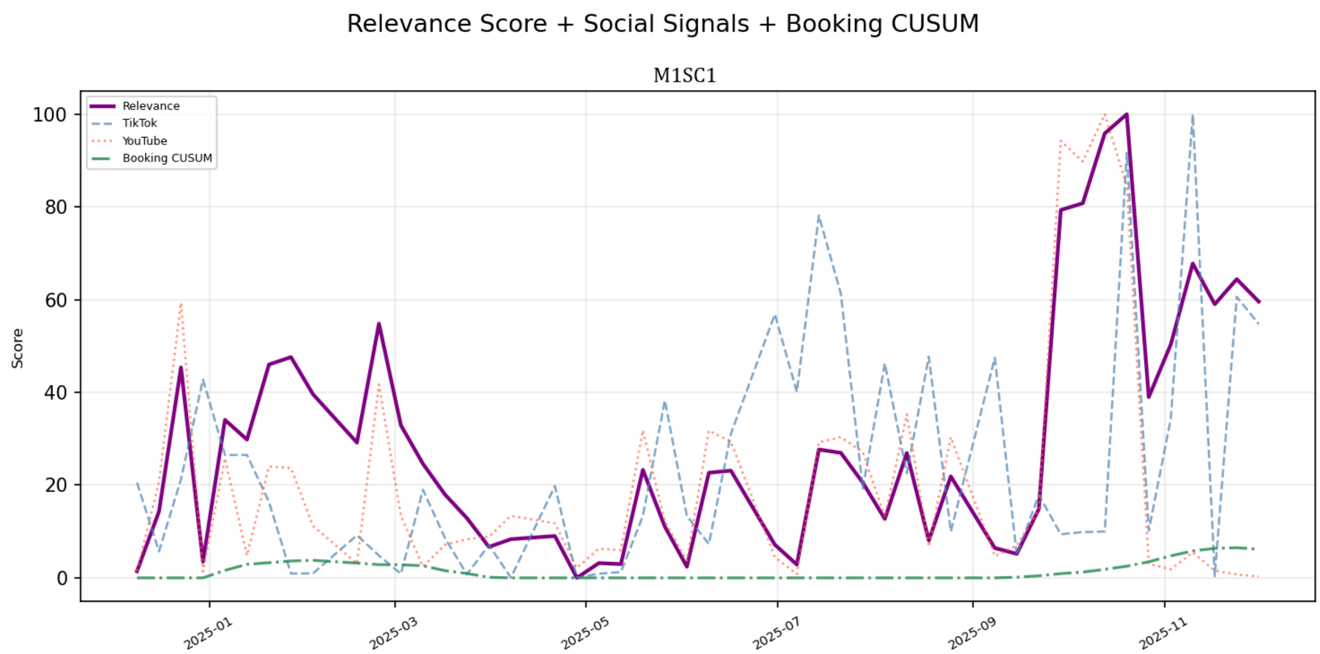

Figure 3: [M1SC1] Relevance Score (purple) overlaid with TikTok (blue dashed), YouTube (red dotted), and Booking CUSUM (green dash-dot). The October 2025 peak (score = 100) is driven by simultaneous TikTok and YouTube spikes. The February 2025 spike is YouTube-only, producing a moderate but not maximum score.

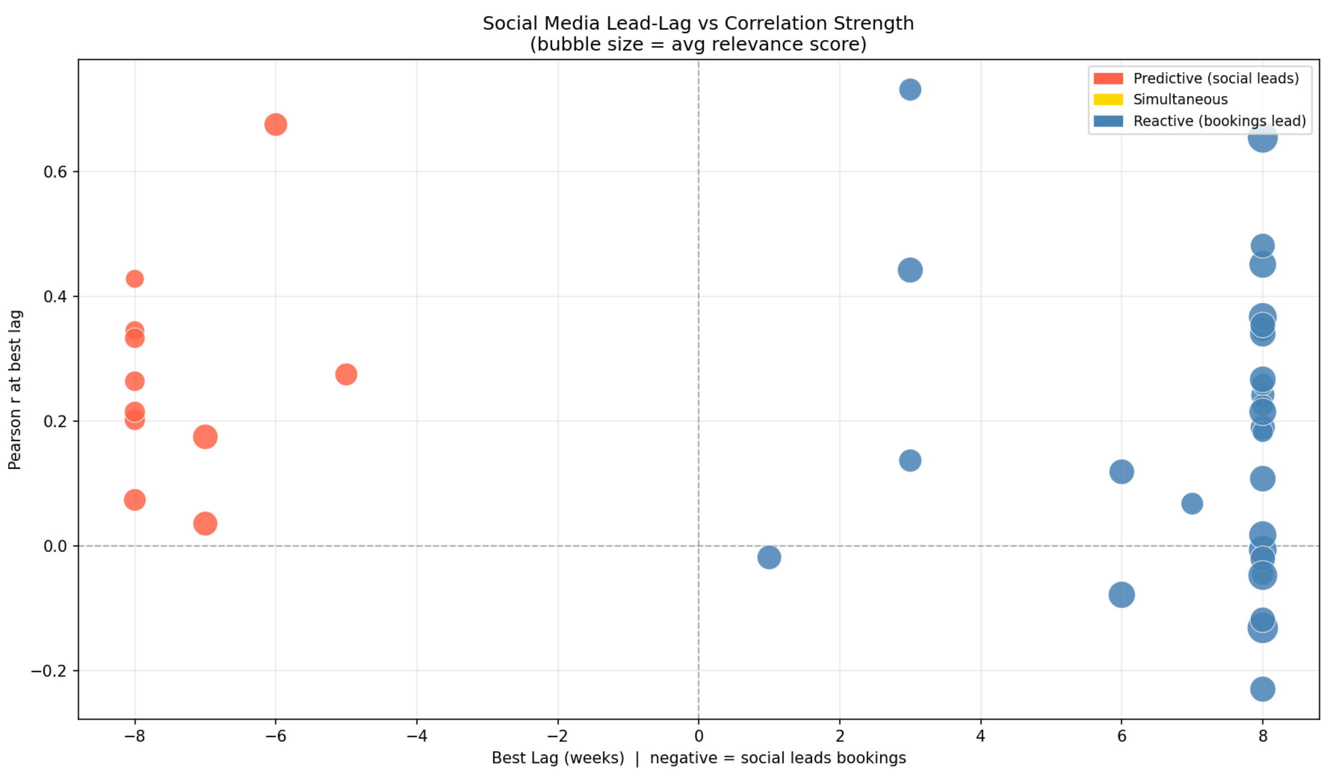

Figure 4: Pearson r vs. optimal lag for all 41 countries. Red points = predictive (social leads bookings); blue = reactive. [M1SC1] (lag $= -6$, $r = 0.628$) is the top predictive country. [M1SC2] (lag $= +3$, $r = 0.459$) is reactive despite high correlation.

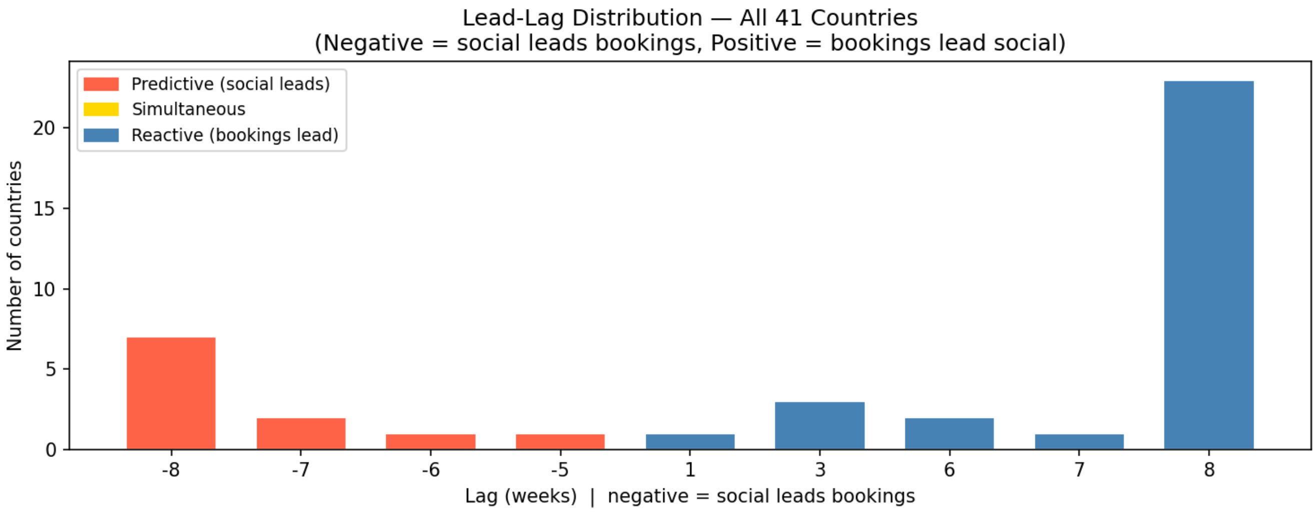

Figure 5: Distribution of optimal lag values. The majority of countries cluster at $\ell = -8$ weeks (social leads by 8 weeks) or $\ell = +8$ weeks (bookings lead by 8 weeks).

Figure 6: Spike-to-surge validation for the top 5 predictive countries. Each panel overlays the social signal (blue), Relevance Score (purple dashed), and actual passengers (green). Green arrows = confirmed predictions; red arrows = false alarms.

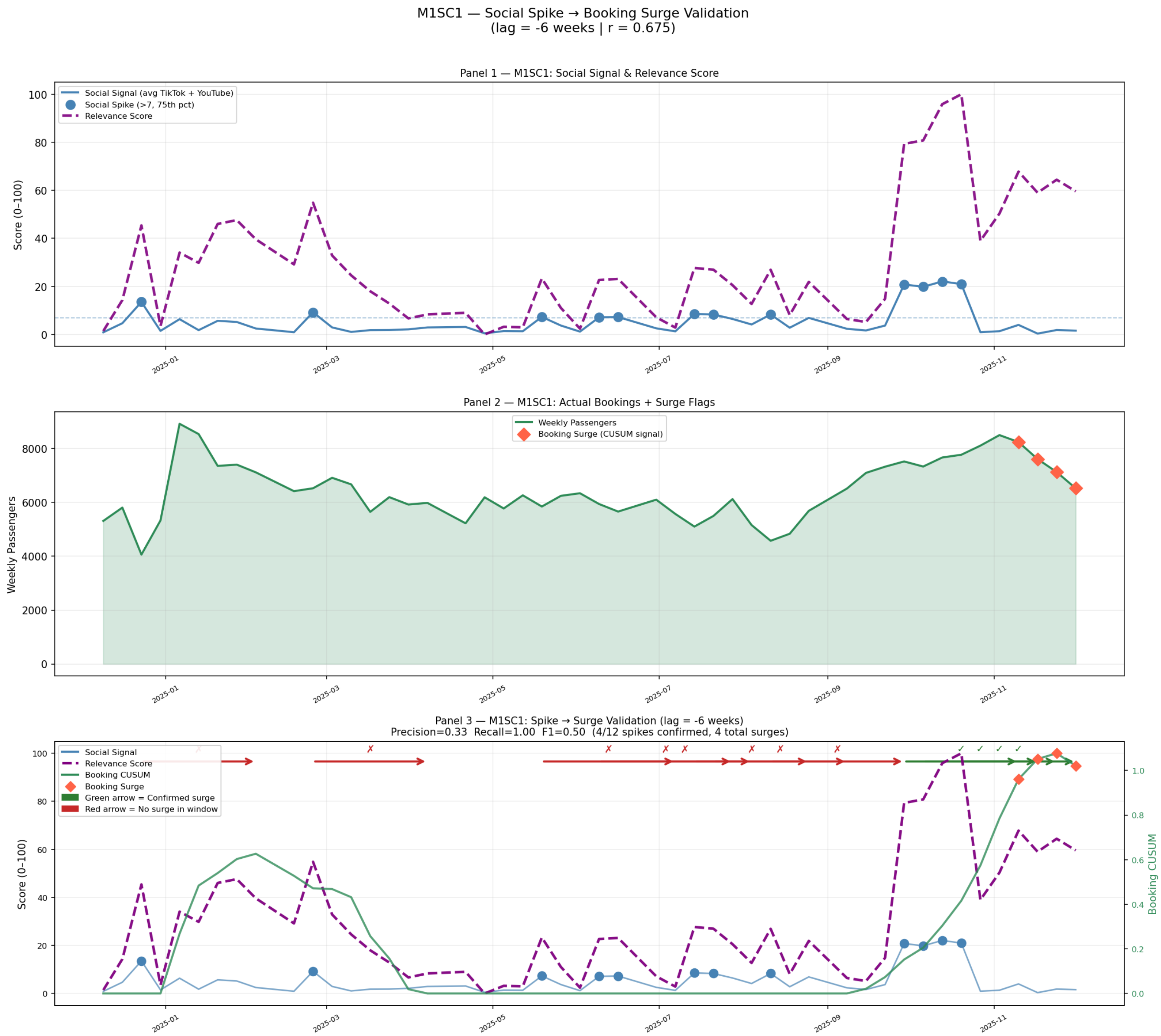

Figure 7: [M1SC1] validation detail. **Panel 1:** social signal (blue) and Relevance Score (purple dashed) with spike markers. **Panel 2:** actual weekly passengers with surge markers (red diamonds). **Panel 3:** prediction test — each arrow connects a social spike to its expected surge 6 weeks later. Green = confirmed; red = false alarm. 4 of 12 spikes confirmed (P=0.333, R=1.000, F1=0.500).

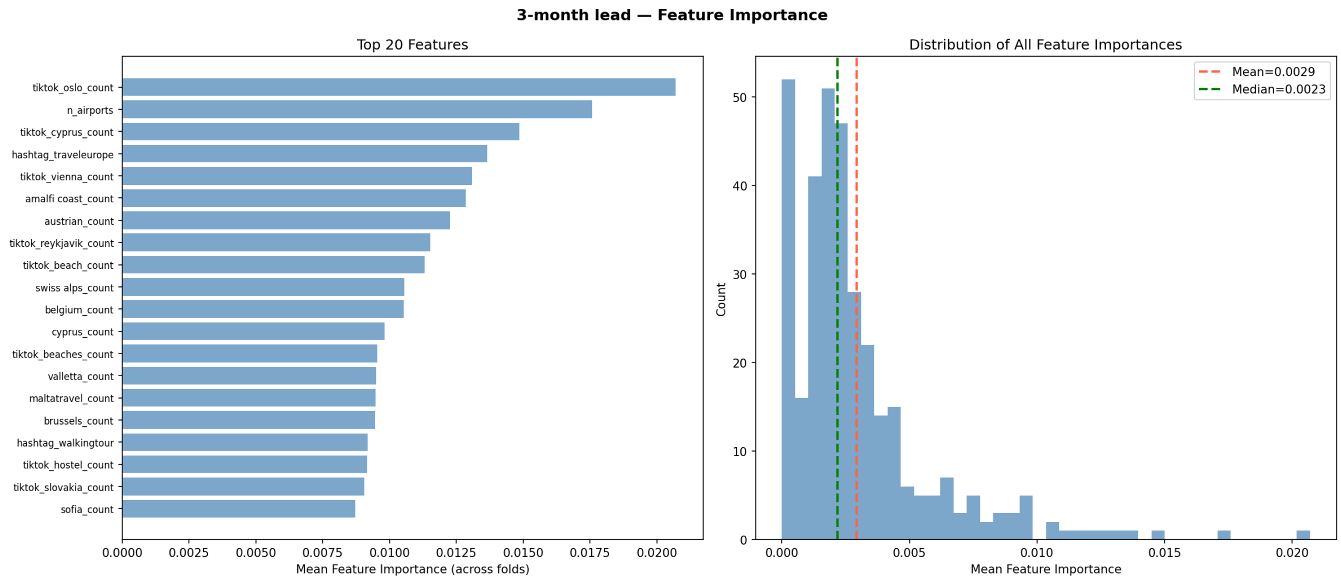

Figure 8: Model 2 overall feature importance.

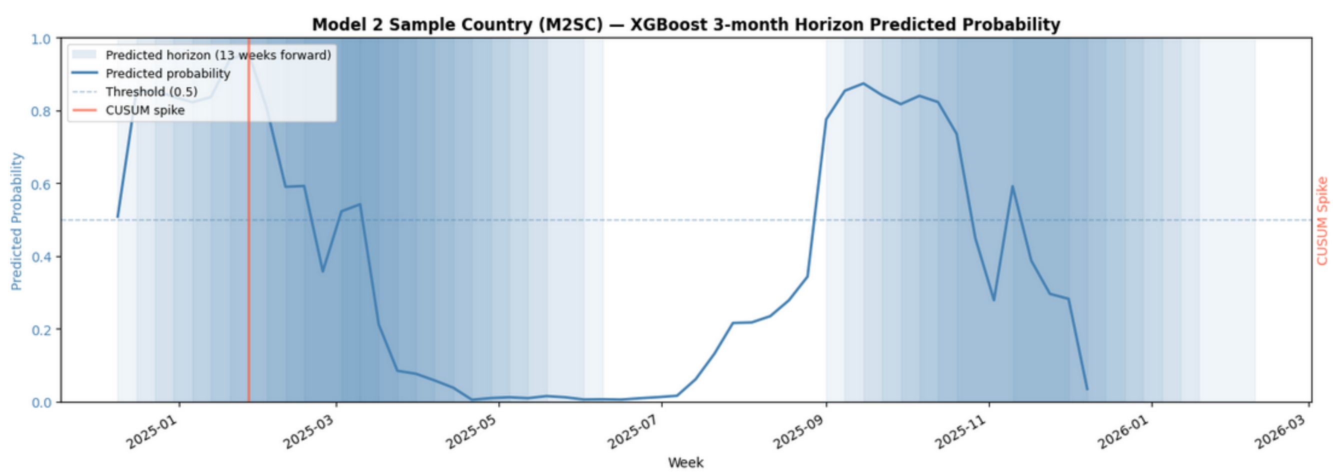

Figure 9: Model 2 analysis for [M2SC].

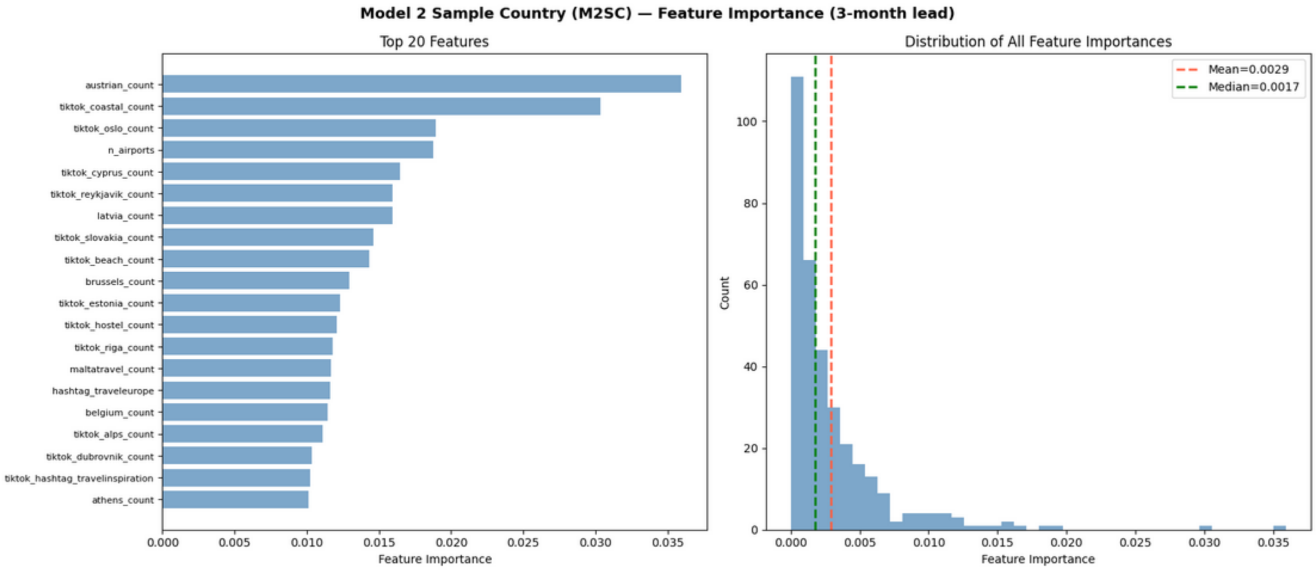

Figure 10: Model 2 feature importance for [M2SC].